$$2E\int_P dA = \frac{\sigma A}{\epsilon_0}$$

The total area of surface either at P or P′

$$\int_P dA = A$$

$$\text{Hence } 2EA = \frac{\sigma A}{\epsilon_0} \text{ or } E = \frac{\sigma}{2\epsilon_0} \qquad (1.70)$$

$$\text{In vector form, } \vec{E} = \frac{\sigma}{2\epsilon_\circ}\hat{n} \qquad (1.71)$$

Here $\hat{n}$ is the outward unit vector normal to the plane. Note that the electric field due to an infinite plane sheet of charge depends on the surface charge density and is independent of the distance *r*.

The electric field will be the same at any point farther away from the charged plane. Equation (1.71) implies that if σ > 0 the electric field at any point P is along outward perpendicular $\hat{n}$ drawn to the plane and if σ < 0, the electric field points inward perpendicularly to the plane ($-\hat{n}$).

For a finite charged plane sheet, equation (1.71) is approximately true only in the middle region of the plane and at points far away from both ends.

(iii) Electric field due to two parallel charged infinite sheets

Consider two infinitely large charged plane sheets with equal and opposite charge densities +σ and -σ which are placed parallel to each other as shown in the Figure 1.39.

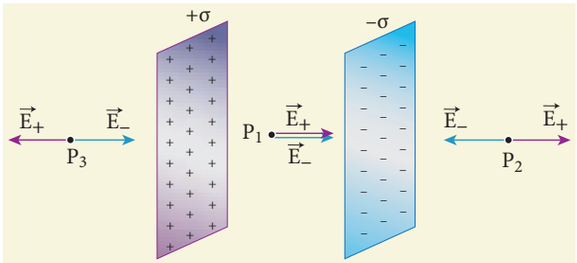


Figure 1.39 Electric field due to two parallel charged sheets

The electric field between the plates and outside the plates is found using Gauss law. The magnitude of the electric field due to an infinite charged plane sheet is $\frac{\sigma}{2\epsilon_\circ}$ and it points perpendicularly outward if σ > 0 and points inward if σ < 0.

At the points P_2 and P_3, the electric field due to both plates are equal in magnitude and opposite in direction (Figure 1.41). As a result, electric field at a point outside the plates is zero. But between the plates, electric fields are in the same direction i.e., towards the right and the total electric field at a point P_1 is

$$E_{inside} = \frac{\sigma}{2\epsilon_\circ} + \frac{\sigma}{2\epsilon_\circ} = \frac{\sigma}{\epsilon_\circ} \qquad (1.72)$$

The direction of the electric field between the plates is directed from positively charged plate to negatively charged plate and is uniform everywhere between the plates.

(iv) Electric field due to a uniformly charged spherical shell

Consider a uniformly charged spherical shell of radius *R* carrying total charge *Q* as shown in Figure 1.40. The electric field at points outside and inside the sphere can be found using Gauss law.

Case (a) At a point outside the shell (*r* > *R*)

Let us choose a point P outside the shell at a distance r from the centre as shown in Figure 1.40 (a). The charge is uniformly distributed on the surface of the sphere (spherical symmetry). Hence the electric field must point radially outward if *Q* > 0 and point radially inward if *Q* < 0. So a spherical Gaussian surface of radius *r* is chosen and

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